



Co-modification by LiAlO₂-coating and Al-doping for LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ as a high-performance cathode material for lithium-ion batteries with a high cutoff voltage



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ABSTRACT

LiAlO₂-coated and Al-doped LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ (Al-modified NCM) materials are successfully synthesized via a facile coprecipitation method. The lattice parameters calculated from X-ray diffraction (XRD) patterns reveal that the slab spacing of the modified material is broadened after Al substitution. A continuous and uniform LiAlO₂ coating layer with a thickness of ~4.3 nm on the surface of Al-doped LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ is confirmed by transmission electron microscopy (TEM). Al-modified NCM shows remarkable improvements in cycling and rate performance and possesses a high capacity retention of 82.06% at 0.5 C after 200 repeated cycles in the voltage range of 3.0–4.6 V, whereas the capacity retention of pristine NCM is only 60.04%. Even at a high current density and an elevated temperature, the Al-modified sample still shows better cycling performance than that of pristine NCM, with capacity retentions of 67.68% (25 °C) and 57.36% (85 °C) after 100 cycles at 10 C. Furthermore, Al-modified NCM delivers a high discharge specific capacity of 142.73 mAh g⁻¹ at a current density of 10 C. The improved electrochemical performance could be ascribed to the synergistic effect of the LiAlO₂ coating and Al doping. The enlarged slab spacing resulting from Al substitution and the Li⁺-conducting LiAlO₂-coating layer benefits lithium ion diffusion. Moreover, the LiAlO₂-coating layer acts as a protective shell and can suppress the increase in impedance during repeated cycling.

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1. Introduction

Recently, the rapid increase in the use of electric vehicles (EVs), hybrid electric vehicles (HEVs), and plug-in hybrid electric vehicles (PHEVs) has hastened the development of energy-storage equipment [1,2]. Lithium-ion batteries (LIBs), one of energy-storage devices, offer many advantages, including high operating voltage, high power and energy densities, long cycle life, low cost, non-toxicity and safety [3–5]. However, LIBs suffer severe capacity fading during repeated cycling and show a poor rate capability [6,7]. New cathode materials have been developed to resolve these problems. Li[Ni_xCo_yMn_z]O₂, a promising lithium containing metal oxide cathode material, has attracted widespread attention because of its inherent merits, such as low cost, safety, high cycling stability

and low toxicity, which are superior over other cathode materials [8]. However, further studies are still needed for large-scale commercialization of this material, especially with respect to its cycling capability in hostile environments, i.e., under high discharge rates, at extreme elevated temperatures and at ultrahigh cutoff voltages.

The structure of the LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ material may change from a layered structure to a spinel structure during repeated cycles under a high cutoff voltage, resulting in poor cycling stability [9–11]. Moreover, when charged up to 4.6 V, electrolyte decomposition and instability of the material surface accelerate capacity fading. Significant efforts have been made to resolve these problems, with bulk doping and surface coating as common modification methods. Bulk doping includes cationic doping (Mo [12], Al [13,14], K [15], Zr [16], Ti [17], Na [18] and V [19]) and anionic doping (F [20,21], Cl [22]). In general, foreign ion doping can alter the valence states of both the oxygen and transition metal through charge compensation mechanisms while changing the distance between the lithium and transition metal layers [16]. Luo et al. [23]

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synthesized a series of Al-doped $\text{Li}_{1-x}\text{Ni}_{0.5}\text{Al}_x\text{Mn}_{1.5-x}\text{O}_4$ cathode materials and found that a small amount of Al improved the rate capability of $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$. Li et al. [24] reported $\text{Li}_{1-x}\text{Al}_x(\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3})_{1-y}\text{Al}_y\text{O}_2$ composites obtained by Al doping that showed an improved rate performance, although the capacity retention was only 85.1% after 100 cycles. To improve the cycling stability, surface coating was employed. As common and promising surface-coating materials, metal oxides (ZrO_2 [25], Al_2O_3 [26–28], MgO , SiO_2 [29], ZnO , TiO_2 [30]) and lithium salts (Li_2SiO_3 [31], Li_2MoO_2 [32], $\text{Li}_2\text{Si}_2\text{O}_5$ [33], Li_2ZrO_3 [34–36], LiAlO_2 [37], LiBO_2 [38]) are being considered. Compared to nonionic conductive Al_2O_3 , LiAlO_2 has greater lithium diffusivity, and the Al_2O_3 coating layer can react with the active material surface to form a Li-containing species such as LiAlO_2 [39]. The capacity retention and rate capability of NCM with an ultrathin LiAlO_2 coating layer are superior to those of NCM with an Al_2O_3 coating layer [40]. Liu et al. [41] reported a LiAlO_2 -coated $\text{Li}_{1.5}\text{Ni}_{0.25}\text{Mn}_{0.75}\text{O}_{2.5}$ material with improved cycling stability and rate capability, but the rate capability of the LiAlO_2 -modified samples was still unsatisfactory (only 88.3 mAh g^{-1} at 10°C). Therefore, single modification by Al doping or a LiAlO_2 coating cannot effectively and simultaneously improve both the rate capability and cycling stability of cathode materials.

Recently, Mo-modified $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$, which combines the Mo doping and a Mo coating, was reported by Zhang et al. [42]. The modified samples afforded a capacity retention of 89.7% after 500 cycles between 3.0 and 4.3 V. Inspired by these studies, we successfully synthesized an Al-doped and LiAlO_2 -coated $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ material by a facile method. In contrast to traditional coating methods, a $\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{CO}_3$ precursor rather than $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ was used for coating with $\text{Al}(\text{OH})_3$. This coating method generates a uniform and continuous coating layer and combines the effects of a LiAlO_2 coating and Al doping for $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$. Moreover, by forming a product that does not need any subsequent treatment steps, the occurrence of chemical deintercalation of lithium is prevented, and the integrity of the material structure is maintained. In this study, the effects of Al modification on the crystal structure, morphology and electrochemical performance of Al-modified NCM are systematically studied by X-ray powder diffraction (XRD), scanning electron microscopy (SEM), transmission electron microscopy (TEM), charge-discharge tests, cyclic voltammetry (CV) and electrochemical impedance

spectroscopy (EIS) techniques. In addition, the morphology and crystal structure of the cycled electrodes are elucidated and revealed in detail.

2. Experimental

2.1. Sample preparation

LiAlO_2 -coated and Al-doped $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ (denoted as Al-modified NCM) was synthesized by a simple coprecipitation method. A schematic diagram of the synthesis processes is shown in Fig. 1. First, stoichiometric amounts of $\text{Ni}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$, $\text{Co}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$, and $\text{Mn}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$ were dissolved in 50 mL of deionized water to form an aqueous solution (1 mol L^{-1}). At the same time, as the precipitant, an aqueous Na_2CO_3 solution was rapidly added into the abovementioned solution. The resulting mixture was vigorously stirred at room temperature for 10 h. Then, the temperature was increased to 80°C and held for 2 h. Simultaneously, appropriate solutions of Na_2CO_3 and $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ were added to the mixture in a dropwise manner to obtain $\text{Al}(\text{OH})_3$ -coated $\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{CO}_3$. The resulting $\text{Al}(\text{OH})_3$ -coated $\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{CO}_3$ was washed, filtered and dried. Finally, the as-synthesized precursor was mixed with lithium carbonate (Li_2CO_3); then, the mixture was sintered at 450°C for 5 h and then at 850°C for 12 h to form different LiAlO_2 -coated and Al-doped $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ materials (1 wt.%, 2 wt.%, and 4 wt.%; Al content in the form of Al_2O_3). For comparison, bare $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ (NCM) was prepared using the same procedure without the addition of solutions of Na_2CO_3 and $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ during the 80°C treatment process.

2.2. Characterization techniques

The phase-structural composition of the as-prepared powders was determined by XRD (Bruker D8 Advance) using $\text{Cu K}\alpha$ radiation in the 2θ range of 10° – 90° with a scan rate of $5^\circ/\text{min}$ and a step size of 0.02° . The detailed morphology of the as-synthesized materials was characterized via SEM (Zeiss Meilin) and TEM (JEM-2100F) instruments equipped with energy dispersive X-ray spectroscopy (EDS).

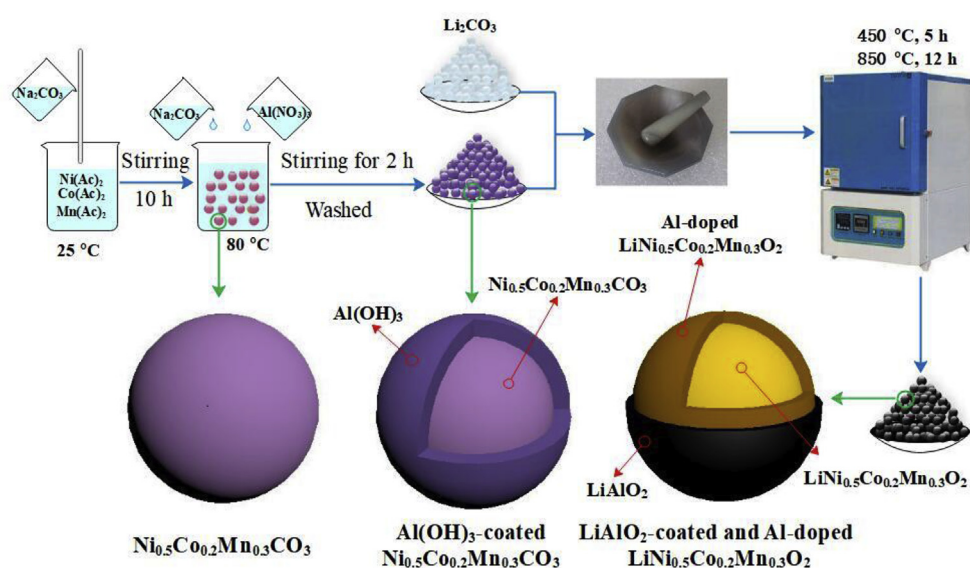


Fig. 1. Schematic diagram of the synthesis process for different Al-modified $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ samples.

2.3. Electrochemical tests

The electrochemical properties were studied by means of two-electrode coin-type half-cells (CR 2016) consisting of a positive electrode (85 wt.% as-prepared active material, 10 wt.% acetylene black and 5 wt.% polyvinylidene fluoride), a Celgard 2400 separator and lithium foil as the counter electrode. The electrolyte was 1 mol L⁻¹ LiPF₆ in ethylene carbonate/dimethyl carbonate/ethyl methyl carbonate (1:1:1, by volume). All cells were assembled in an argon-filled glove box in which both the oxygen and moisture contents were controlled to be less than 1 ppm. Galvanostatic charge/discharge tests were conducted on a BTS-XWJ-6 testing system (Neware, China). All cells were aged for 12 h before testing and then cycled in two different voltage ranges (3–4.3 V and 3–4.6 V (vs. Li/Li⁺)) at the desired C rate (1C = 160 mA g⁻¹). CV measurements were carried out with an electrochemical workstation (CHI660C, Shanghai Chenhua) at a scan rate of 0.1 mV s⁻¹. EIS was also measured over the frequency range from 0.01 Hz to 100 kHz with an amplitude of 5 mV.

3. Results and discussion

A reaction procedure with only Na₂CO₃ and Al(NO₃)₃ under similar synthetic conditions as those used for Al-modified NCM was performed to confirm the possible formation of LiAlO₂ on the surface of NCM. As shown in Fig. 2 (a), all peaks of the reaction product (denoted as as-prepared LiAlO₂) of Na₂CO₃ and Al(NO₃)₃ can be indexed to the γ -LiAlO₂ phase (JCPDS No. 38-1464), while reflections corresponding to Al₂O₃ were not observed, confirming that the coating layer on NCM was composed of LiAlO₂, which was as expected since the same procedure was used. Fig. 2 (b) shows the XRD patterns of bare and Al-modified NCM. All peaks are well indexed to the layered hexagonal α -NaFeO₂ structure with the space group R-3m. Distinct splitting of (006)/(102) and (108)/(110) for all samples indicates that a well-ordered layered structure formed. No impurities or secondary phases can be observed in the XRD patterns due to the low content of the LiAlO₂ coating. The insert image in Fig. 2 (b) is an enlarged view of the (003) peak. As the Al content increased, the (003) peak migrated to lower 2 θ values, suggesting that Al was successfully incorporated into the crystal lattice [26]. To further explore the effects of Al modification on the lattice structure, the lattice parameters of the as-prepared materials were determined and are listed in Table 1. The values of *c* and *a* increased after the addition of the Al element. In addition, the *c/a* ratios of all the as-prepared materials are greater than 4.9, showing that the materials are layered oxides. The values of *I*₍₀₀₃₎/*I*₍₁₀₄₎

Table 1

Lattice parameters of bare NCM and different Al-modified NCM samples.

Sample	<i>a</i> (nm)	<i>c</i> (nm)	<i>c/a</i>	<i>I</i> ₍₀₀₃₎ / <i>I</i> ₍₁₀₄₎	<i>R</i>
0 wt.%	0.28727	1.41308	4.9190	1.2662	0.5907
1 wt.%	0.28686	1.42251	4.9589	1.3685	0.5914
2 wt.%	0.28684	1.42554	4.9698	1.3303	0.4795
4 wt.%	0.28717	1.41785	4.9373	0.9255	0.7415

I₍₁₀₄₎ for all samples are greater than 1.2, indicating low cation disorder in the crystal lattice, whereas 4 wt.% Al-modified NCM shows severe cation disorder because of the reaction 2Al(OH)₃+Li₂CO₃→2LiAlO₂+CO₂+3H₂O at elevated temperatures. Among all the samples, 2 wt.% Al-modified NCM has the largest value of *c/a* (4.9698), a higher value of *I*₍₀₀₃₎/*I*₍₁₀₄₎ (1.3303, >1.2) and the smallest value of *R* (0.4795, as calculated from *R*=(*I*₍₀₀₆₎+*I*₍₁₀₂₎)/*I*₍₁₀₁₎), suggesting that this sample would exhibit better electrochemical performance. Therefore, the following discussions are focused on 2 wt.% Al-modified NCM and bare NCM.

To investigate the effects of Al modification on the morphology of the samples, SEM study was conducted, and the results are presented in Fig. 3. As shown in the figures, both bare and Al-modified NCM are composed of irregular primary particles with sizes ranging from 200 to 700 nm. A similar particle surface can be observed on both samples because of the thinly coated LiAlO₂ layer. Some of the primary particles of bare NCM agglomerated after high-temperature calcination (with a maximum secondary particle size of ~2.5 μ m), whereas the Al-modified NCM did not agglomerate, probably due to the presence of Al(OH)₃. The Al(OH)₃ coatings on the surface of the precursor can effectively prevent the primary particles from agglomeration. The corresponding EDS results are given in Fig. 3 (e) and (f). Both bare and Al-modified NCM have a similar ratio of Ni:Co:Mn that is approximately equal to 5:2:3 (owing to the inaccuracy of the O contents), meaning that the LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ material was successfully synthesized. In addition, the Al content of Al-modified NCM was nearly equal to the theoretically calculated value.

To verify the formation of LiAlO₂ coatings on the surface of Al-modified NCM powder, TEM images of bare and Al-modified NCM samples are presented in Fig. 4 (a) and (b). Compared to bare NCM with a smooth surface, Al-modified NCM has a uniform LiAlO₂ coating layer on the surface of LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ with a thickness of approximately 4.3 nm. The d-spacing of the lattice fringes for the material in Fig. 4 (a) is 0.2069 nm, corresponding to the (104) plane of the hexagonal layered phase. Two sets of lattice fringes are present in Fig. 4 (b): the lattice fringes of 0.2034 nm (corresponding

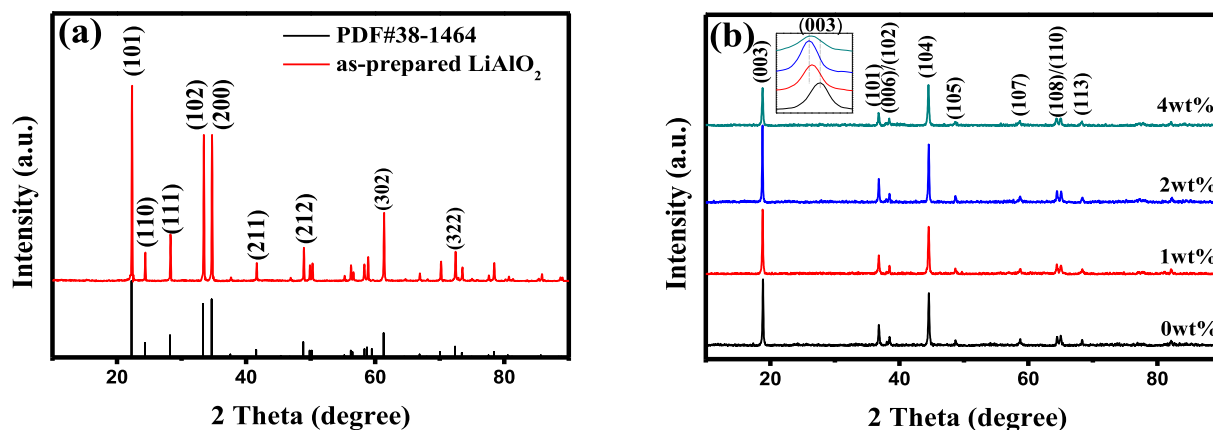


Fig. 2. XRD patterns of the as-prepared samples: (a) as-prepared LiAlO₂, (b) bare NCM and different Al-modified NCM samples.

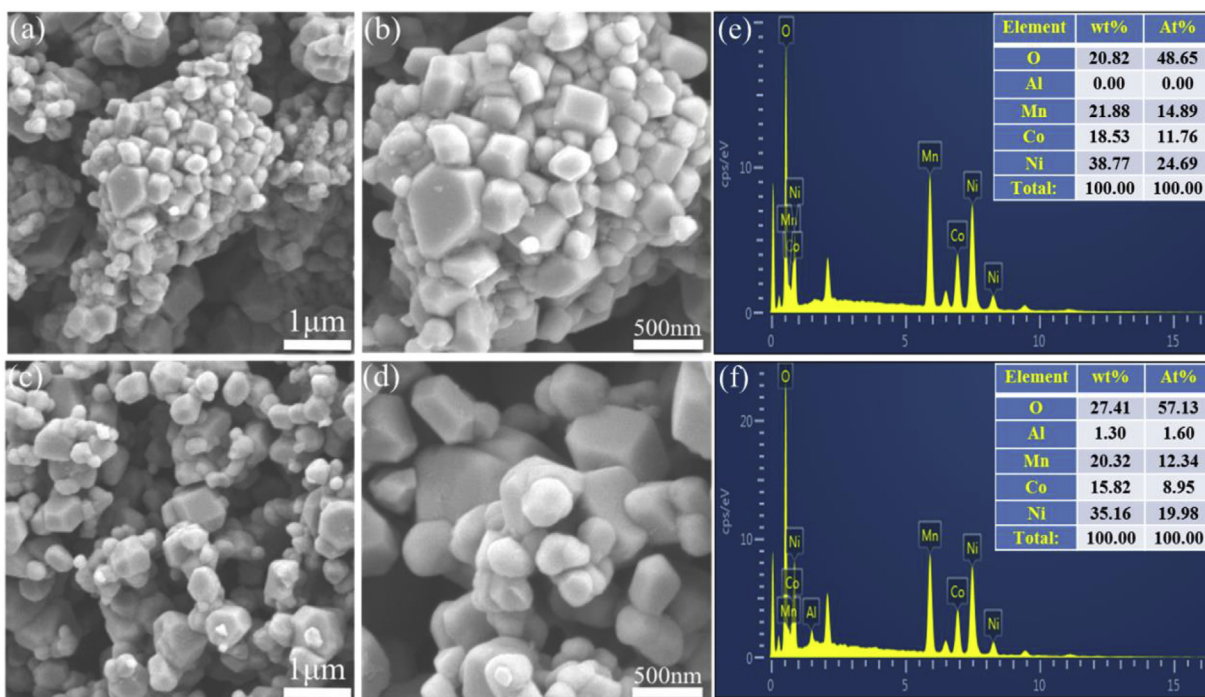


Fig. 3. SEM images of (a, b) bare NCM, (c, d) Al-modified NCM, and the EDS results of (e) bare NCM, and (f) Al-modified NCM.

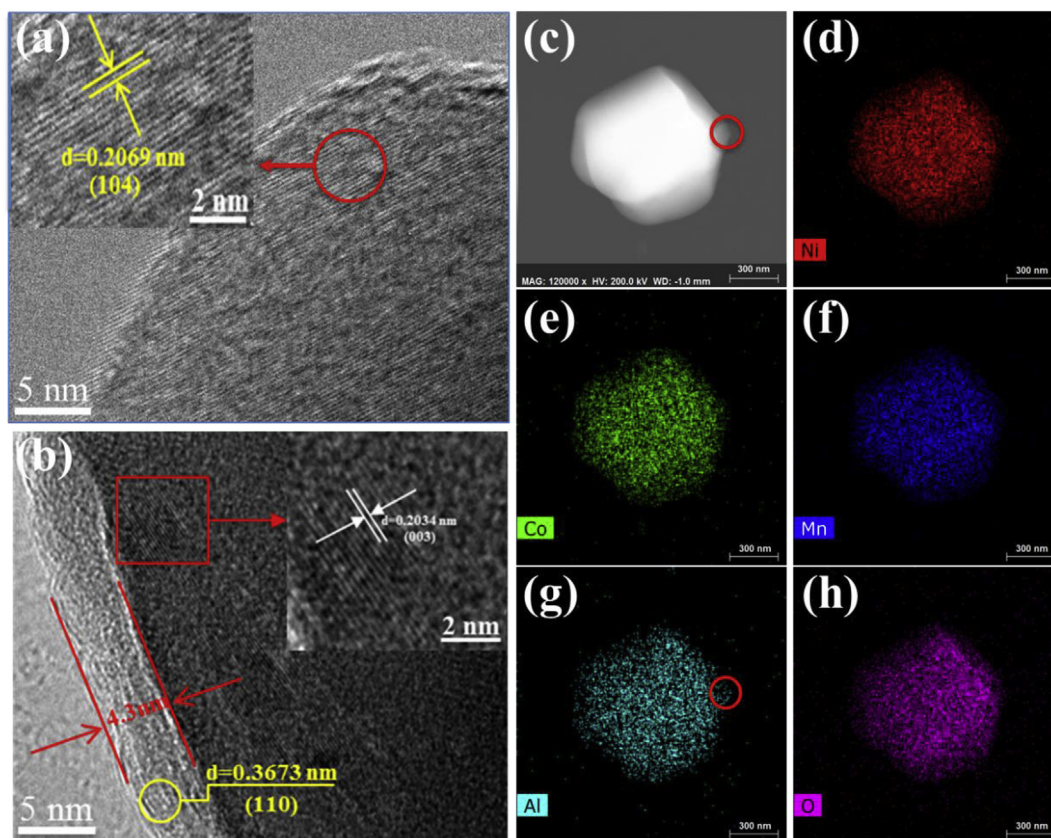


Fig. 4. TEM images of (a) bare NCM, (b, c) Al-modified NCM, and EDS mapping images of (d) Ni, (e) Co, (f) Mn, (g) Al, and (h) O.

to the (003) plane of the hexagonal layered phase) and the lattice fringes of 0.3673 nm (corresponding to the (110) plane of LiAlO_2). EDS analysis of Al-modified NCM was performed, and the results are shown in Fig. 4(d–h). As the results show, all elements are uniformly distributed on the particles. A uniform distribution of the Al element indicates a uniform LiAlO_2 coating layer on the surface of Al-modified NCM particles. All the above evidences confirm that Al-doped and LiAlO_2 -coated $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ was successfully synthesized.

Fig. 5 (a) presents the initial charge and discharge curves for bare and Al-modified NCM at 0.5 C in the voltage range of 3.0–4.6 V. Both curves displayed typical electrochemical behavior with a smooth and monotonous voltage plateau ascribed to lithium-ion extraction from the $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ material. Bare NCM shows initial charge and discharge specific capacities of 238.22 mAh g^{-1} and 185.25 mAh g^{-1} , respectively, and the corresponding coulombic efficiency is 77.8%. The initial charge and discharge specific capacities of Al-modified NCM are 228.22 mAh g^{-1} and 183.43 mAh g^{-1} , respectively, which are slightly lower than those of bare NCM due to the inactive LiAlO_2 coating. However, Al-modified NCM gives a higher coulombic efficiency of 80.4% because the existence of a LiAlO_2 coating layer can suppress the oxidative decomposition of the electrolyte components on a high-voltage cathode.

The cycling performance of different Al-modified NCM samples at 2 C is shown in Fig. S2, while the cycling performance of the as-prepared material at 0.5 C over two different voltage ranges of 3–4.3 V and 3–4.6 V is given in Fig. 5 (b) and (c). According to Fig. 5 (b), bare and Al-modified NCM exhibit initial discharge specific

capacities of 158.98 mAh g^{-1} (initial coulombic efficiency: 83.9%) and 152.26 mAh g^{-1} (initial coulombic efficiency: 85.3%). Both samples have good cycling performance owing to the narrow voltage range, especially the Al-modified sample (99.42% capacity retention after 100 cycles). When the cutoff voltage reached 4.6 V, both bare and Al-modified NCM produced higher initial discharge specific capacities of 185.24 mAh g^{-1} and 183.43 mAh g^{-1} , respectively. However, the electrolyte was probably decomposing at such a high cutoff voltage, resulting in poor cycling performance. After 200 cycles, bare NCM delivered a discharge specific capacity of only 111.17 mAh g^{-1} (60.04% capacity retention), whereas Al-modified NCM possessed a higher discharge specific capacity of 150.57 mAh g^{-1} (82.08% capacity retention). This capacity retention is much better than that in other reports, for example, 75.5% capacity retention after 100 cycles [26] and 85% capacity retention after 100 cycles [43]. The improved cycling performance can be attributed to the LiAlO_2 coatings, which can protect the electrode from HF attack. As shown in Fig. S3, LiAlO_2 coatings can also effectively reduce voltage fading during cycling.

Fig. 5 (d) shows the rate performance of bare and Al-modified NCM over 3–4.6 V, and a comparison of the rate performances for all the as-prepared materials is presented in Fig. S4. As the current density increases, the two samples show a strong decay in capacity because of the electrode polarization. Al-modified NCM delivers an extremely high discharge specific capacity of 142.73 mAh g^{-1} at 10 C, which is much higher than the discharge specific capacity of 98.46 mAh g^{-1} for bare NCM. When the current density returns to 0.1 C, bare and Al-modified NCM deliver discharge specific capacities of 190.81 mAh g^{-1} and 192.87 mAh g^{-1}

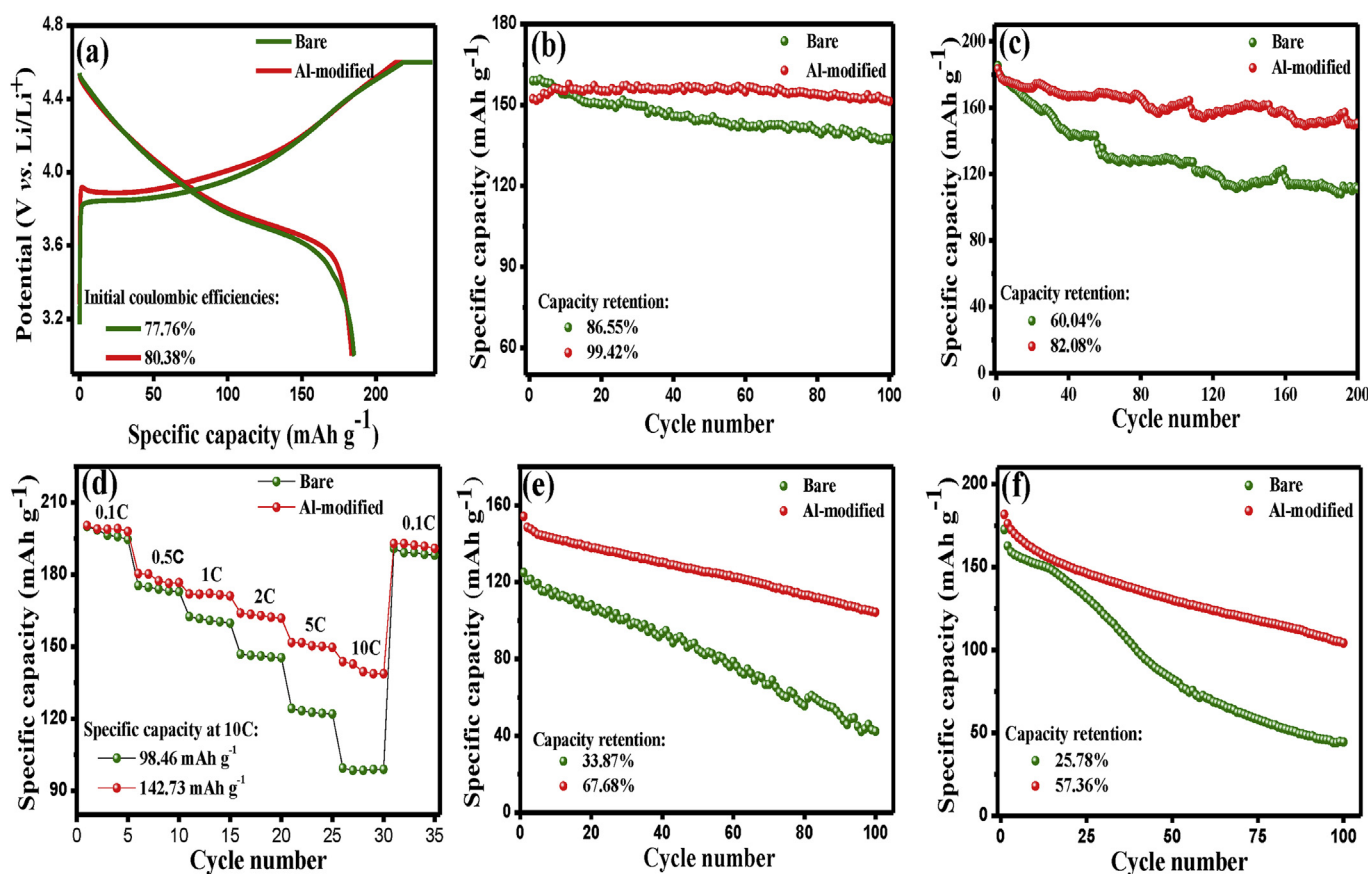


Fig. 5. (a) Initial charge and discharge curves of the as-prepared samples at 0.5 C. Cycling performance of the as-prepared materials at 0.5 C (b) at 3–4.3 V and (c) at 3–4.6 V. (d) Rate performance of the as-prepared samples. Cycling performance at 10 C at different temperatures of (e) 25 °C and (f) 85 °C.

g^{-1} , respectively, meaning that both samples can undergo high current density charging and discharging. The rate performance of Al-modified NCM is much better than that described in other reports [26,43]. The outstanding rate performance of Al-modified NCM is ascribed to the following reasons: i) the $\text{Al}(\text{OH})_3$ coating on the precursor surface can effectively prevent primary particles from agglomeration during high-temperature calcination; ii) the enlarged slab spacing due to Al doping is beneficial to lithium ion transport; and iii) the LiAlO_2 coating layer is a Li^+ -conducting layer.

The cycling performance of bare and Al-modified NCM at 25°C and 85°C at 10 C is presented in Fig. 5 (e) and (f), respectively. In Fig. 5 (e), bare NCM possesses an initial discharge specific capacity of $124.98 \text{ mAh g}^{-1}$, but this value rapidly decreases to 42.34 mAh g^{-1} after 100 cycles (33.87% capacity retention), whereas Al-modified NCM delivers an initial discharge specific capacity of $154.15 \text{ mAh g}^{-1}$ and maintains a value of $104.32 \text{ mAh g}^{-1}$ after 100 cycles (67.68% capacity retention). As shown in Fig. 5 (f), when the temperature increases to 85°C , Al-modified NCM delivers a discharge specific capacity of $104.19 \text{ mAh g}^{-1}$ after 100 cycles (57.36% capacity retention). However, bare NCM delivers a discharge specific capacity of only 44.50 mAh g^{-1} (25.78% capacity retention). Therefore, Al-modified NCM shows better cycle stability than bare NCM at both room temperature and elevated temperature. The improved cycling performance at a high current density and elevated temperature indicates that Al-modified NCM can be used under extreme conditions.

Fig. 6 presents the cyclic voltammograms of bare and Al-modified NCM samples in the voltage range of 3.0–4.6 V with a scan rate of 0.1 mV s^{-1} for different cycles. As shown in the figures,

all CV curves show only one redox couple, suggesting that no phase transition occurs during the extraction/insertion of lithium ions. Due to the formation of a solid electrolyte interface (SEI) or the occurrence of side reactions, the anodic and cathodic peaks show a potential difference (ΔV). In addition, the value of ΔV represents the degree of electrochemical reversibility [44]. Cyclic voltammograms of the as-prepared samples over the first three cycles are shown in Fig. 6 (a) and (b). The value of ΔV for bare NCM is 0.344 V , whereas that of Al-modified NCM is smaller at approximately 0.138 V , revealing that the reversibility and reactivity of the Al-modified sample was improved by Al modification. This result is in accordance with properties of Al-modified NCM having a higher first columbic efficiency. After the cycling, the value of ΔV for bare NCM increased from 0.109 V to 0.251 V , whereas that of Al-modified NCM only increased from 0.093 V to 0.198 V , meaning that Al modification can alleviate polarization and reduce the occurrence of side reactions, which is conducive to improving the cycle performance of materials.

To investigate the beneficial effects of Al modification on the high-voltage electrochemical properties of $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$, EIS was conducted at a charge state of 4.6 V after different cycles at room temperature. As shown in Fig. 7 (a) and (b), all curves consist of two semicircles and a sloped line. In general, the high-to-medium frequency semicircle denotes the SEI resistance (R_{SEI}), and the low-frequency semicircle reflects the charge-transfer resistance (R_{ct}). The slope of the line in the low-frequency region corresponds to the Warburg impedance (W). All obtained EIS spectra were simulated by the equivalent circuit displayed in Fig. 7 (a), and the fitting resistance values are listed in Table 2. As shown

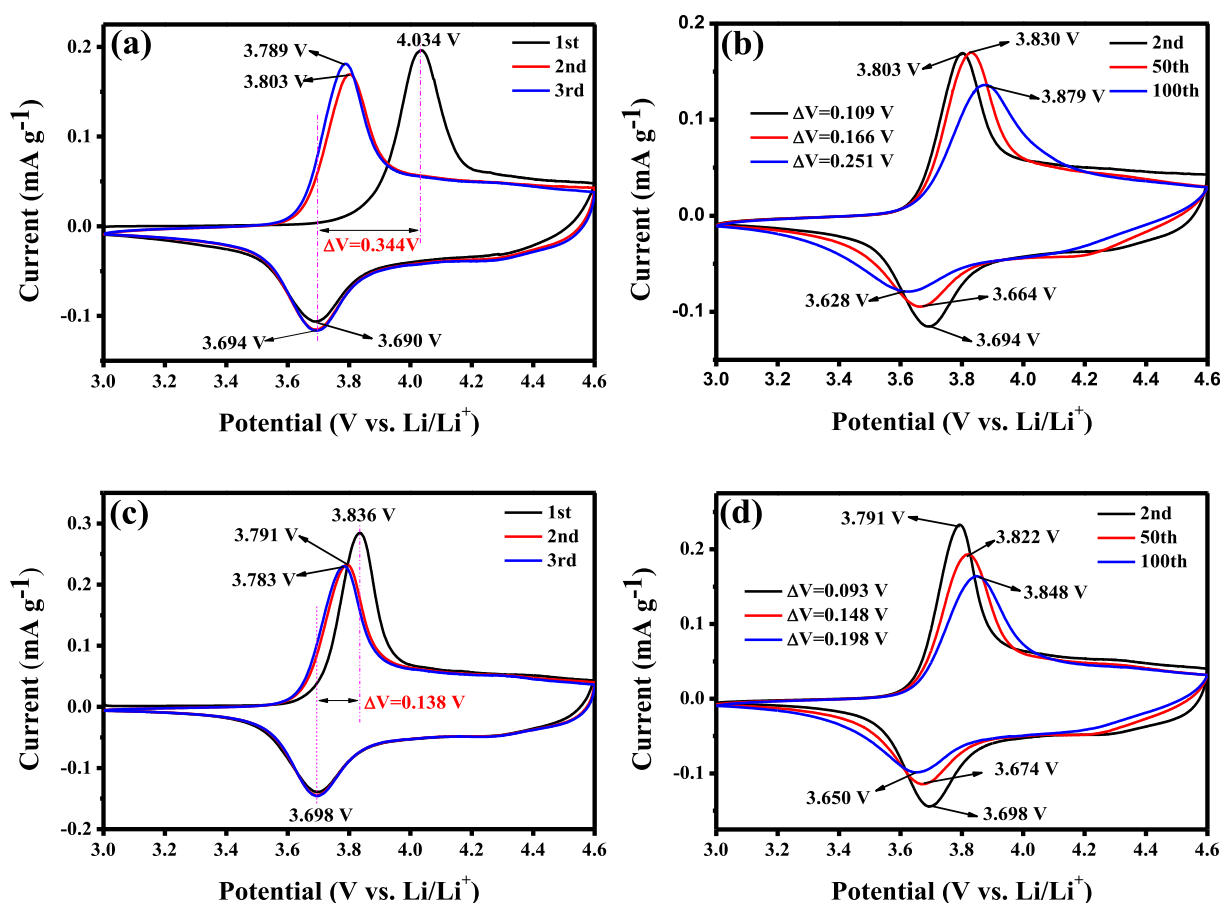


Fig. 6. Cyclic voltammograms of (a, b) bare NCM and (c, d) Al-modified NCM at 0.1 mV s^{-1} between 3.0 and 4.6 V for different cycles.

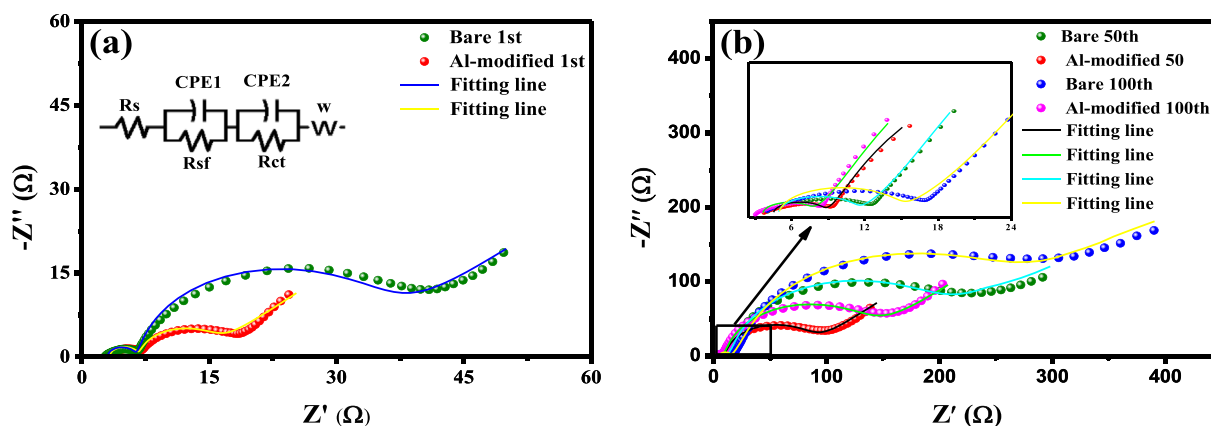


Fig. 7. Nyquist plots of the as-prepared materials after different cycles (a) 1st, (b) 50th and 100th.

Table 2

Impedance parameters of bare NCM and Al-modified NCM electrodes calculated from the equivalent circuit.

Sample	R_s (Ω)	R_{SEI} (Ω)	R_{ct} (Ω)
Bare	2.989	3.105	25.240
Al-modified	3.228	3.216	11.615
Bare-50 th	4.883	6.650	198.400
Al-modified-50 th	4.485	4.247	83.31
Bare-100 th	4.946	10.298	250.984
Al-modified-100 th	3.605	4.079	139.135

in Table 2, the electrolytic resistances R_s in the cells of bare and Al-modified NCM are similar and very small. The R_{ct} of bare NCM is 25.24 Ω for the first cycle and increases to 198.40 Ω after 50 cycles (further increasing to 250.98 Ω after 100 cycles), whereas the R_{ct} of Al-modified NCM increases from 11.62 Ω for the first cycle to a value of only 83.31 Ω after 50 cycles (further increasing to 139.15 Ω after 100 cycles). This analysis implies that the LiAlO_2 coating layer can suppress the increase in impedance during cycling by reducing the occurrence of side reactions between the electrodes and electrolyte.

To further investigate how Al modification affects the rate performance of NCM, the Li^+ diffusion coefficient was calculated according to the following equation [45]:

$$D_{\text{Li}^+} = \frac{R^2 T^2}{2A^2 n^4 F^4 C^2 \sigma^2}$$

where R is the gas constant (8.314 J mol⁻¹ K⁻¹), T is the absolute temperature (298.15 K), A is the surface area of the electrode, n is the number of electrons transferred according to the reaction of lithium-ion insertion or extraction, F is the Faradic constant (96500 C mol⁻¹), c is lithium ion concentration in the cathode, and σ is the Warburg factor. The Warburg factor σ can be calculated based on the following equation:

$$Z' = R_D + R_L + \sigma \omega^{-1/2}$$

The linear relationships between the real impedance Z' and $\omega^{-1/2}$ in the low-frequency region are shown in Fig. 8. The calculated D_{Li^+} values of both samples after different cycles are shown in Table 3. As illustrated in Table 3, Al-modified NCM possesses a high value of $4.720 \times 10^{-10} \text{ cm}^2 \text{ s}^{-1}$ in the first cycle, and this value decreases to $1.04 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ after 50 cycles (further decreasing to $4.965 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$ after 100 cycles). However, the value of bare NCM is only $8.372 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ in the first cycle and rapidly

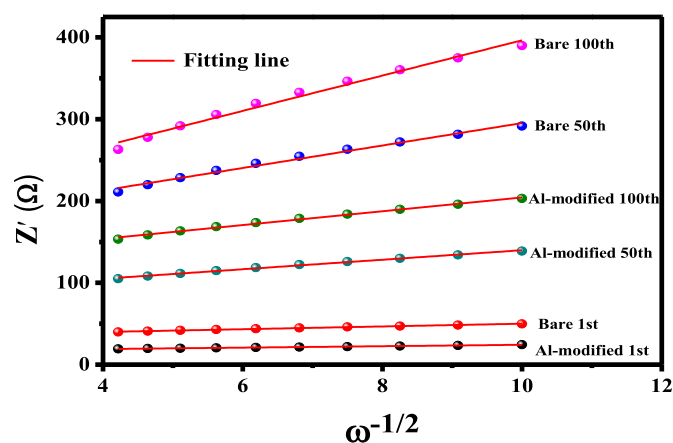


Fig. 8. Relationship plot for Z' and $\omega^{-1/2}$ at a low frequency.

Table 3

Tap density and Li^+ diffusion coefficients of bare NCM and Al-modified NCM.

Sample	Tap density (g cm^{-3})	σ ($\Omega \text{ cm}^2 \text{ s}^{1/2}$)	D_{Li^+} ($\text{cm}^2 \text{ s}^{-1}$)
Bare-1st	1.93	1.67999	8.372×10^{-11}
Bare-50 th		13.72992	1.253×10^{-12}
Bare-100 th		21.59159	5.069×10^{-13}
Al-modified-1st	1.58	0.86427	4.720×10^{-10}
Al-modified-50 th		5.82137	1.040×10^{-11}
Al-modified-100 th		8.42665	4.965×10^{-12}

decreases to $5.069 \times 10^{-13} \text{ cm}^2 \text{ s}^{-1}$ after 100 cycles. The D_{Li^+} values of both samples decreased to a certain extent due to the formation of an SEI and the occurrence of side reactions between the electrodes and electrolyte. However, Al-modified NCM always shows a higher Li^+ diffusion coefficient than that of bare NCM. Therefore, the higher Li^+ diffusion coefficient and the lower charge-transfer impedance result in an improved rate performance.

The structural stabilities of bare and Al-modified NCM after 100 cycles at 10 C under a high cutoff voltage were also investigated; the results are shown in Fig. 9 (a) and (b), and the corresponding EDS mapping is presented in Fig. S5. After cycling at a high current density, bare NCM particles suffer from severe damage and collapse into small fragments. However, compared with bare NCM particles, Al-modified NCM particles have only small sections that collapse into fragments, meaning that Al-modified NCM retains most of the large particles. This result confirms that the existence of the LiAlO_2

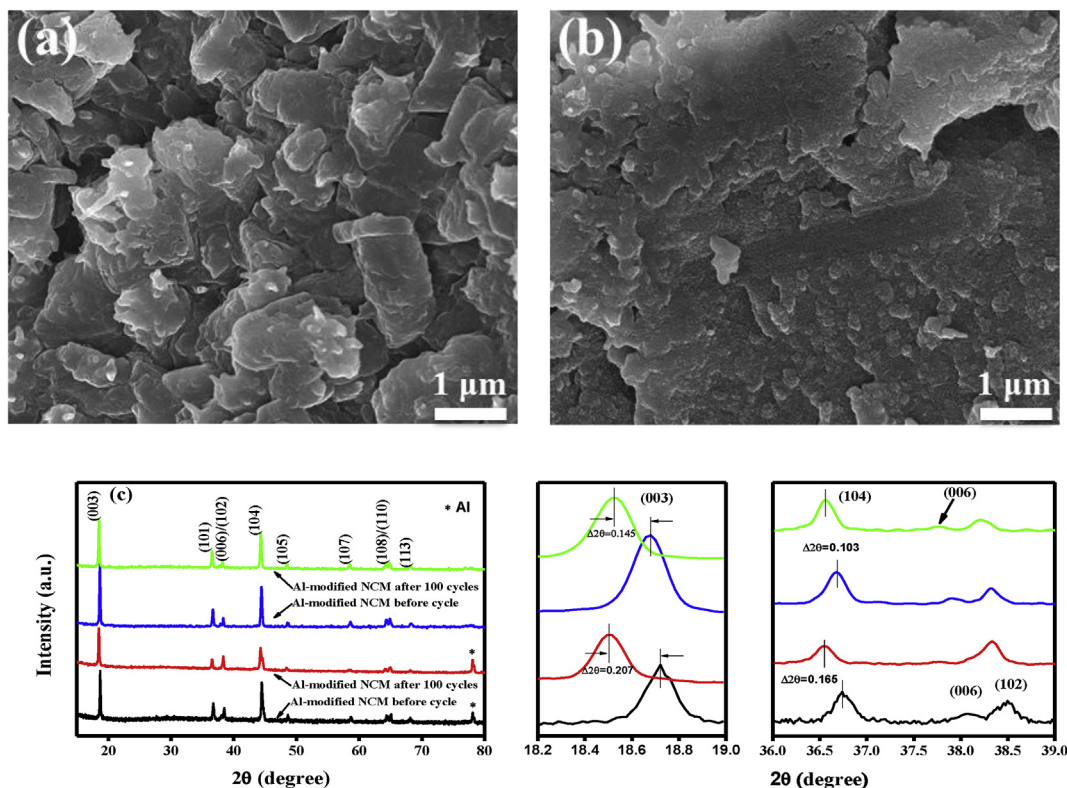


Fig. 9. SEM images of (a) Al-modified NCM, (b) bare NCM electrodes after 100 cycles at 10 C over 3.0–4.6 V and (c) comparison of the XRD patterns for bare and Al-modified NCM electrodes before cycling and after 100 cycles.

coating on the particle surface can successfully prevent particles from direct exposure to the electrolyte in such a way that the structure is not easily destroyed. Hence, Al-modified NCM delivers better cycling stability. Furthermore, the XRD patterns of the cycled bare and Al-modified NCM electrodes are shown in Fig. 9 (c). Compared with bare NCM, Al-modified NCM exhibits a smaller shift of the (003) peak and a more obvious (006) peak. This result confirms that Al-modified NCM has better structural stability than that of pristine NCM, ensuring its superior high-voltage electrochemical performance.

4. Conclusions

LiAlO₂-coated and Al-doped LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ materials with a high rate capability and excellent cycling performance were successfully synthesized via a simple coprecipitation method. The effects of Al modification on the crystal structure, morphology and electrochemical performance for Al-modified NCM were systematically studied. Due to the comodification by LiAlO₂-coating and Al-doping, Al-modified NCM exhibits a high capacity retention of 82.06% at 0.5 C after 200 cycles in the voltage range of 3–4.6 V. Even at an elevated temperature (85 °C), the modified material still exhibits a capacity retention of 57.36% at 10 C after 100 cycles. Additionally, Al-modified NCM delivers a high discharge specific capacity of 142.73 mAh g⁻¹ at 10 C due to the enlarged slab spacing of the crystal structure and the superior lithium-ion conductivity of the LiAlO₂ coating, whereas the discharge specific capacity of bare NCM is only 98.46 mAh g⁻¹. The Al modification can effectively inhibit the collapse of the material structure and reduce the increase in impedance during cycling by reducing the occurrence of side reactions between the electrodes and electrolyte.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jallcom.2018.07.223>.

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